

Part A · Multiple Choice

2 points each

1. **a) 24,000 units on the ground**

p. 52 — the representative fraction is accurate regardless of the unit used: 1 inch to 24,000 inches, 1 cm to 24,000 cm, and so on.

2. **d) It remains true when the map is shrunk or magnified**

p. 53 — because the bar shrinks and grows along with the map itself, a photocopied or resized map keeps a correct graphic scale.

3. **a) a steep slope**

p. 55 — widely spaced contours (or none at all) mean the terrain is relatively flat. Every fifth line is a darker, labeled index contour.

4. **b) preserves shape and angle but strongly distorts area**

p. 60 — the classic illustration is Greenland, which is only 7% of the land area of Africa but appears just as large on a Mercator map.

5. **c) a plane**

p. 58 — cylinders are used near the equator (great circles), cones for the mid-latitudes (small circles), and planes for the poles (points).

6. **b) 6 miles on a side**

p. 64 — 6 miles on a side gives 36 square miles, which is then subdivided into 36 sections of 1 square mile each. A quarter-quarter section is the familiar 40-acre parcel.

Part B · True or False

1 point each

7. **FALSE**

That is resolution. Extent describes the area visible on the map — how far the map is zoomed out. (p. 55)

8. **TRUE**

pp. 59–60 — because flattening a sphere always involves shearing, tearing, or compression, the mapmaker must decide which property matters most for the map's purpose.

Part C · Short Answer

3 points each

9. **Shearing, tearing, and compressing. Developable surfaces eliminate tearing.**

Rubric: 1.5 pts for naming all three (0.5 each). 1 pt for identifying tearing as the one eliminated. 0.5 pt for the reason: tearing is considered the worst of the three because it would leave the map full of holes. Strong answers note that developable surfaces (plane, cone, cylinder) still produce shearing and compression, and that interrupted projections such as Goode Homolosine deliberately accept tearing in exchange for less shape and area distortion. (pp. 57–58)

10. Conformal projections preserve shape and angle but distort area; equal-area projections preserve relative size but distort shape.

Rubric: 1 pt each definition; 1 pt for reasonable uses. Conformal (e.g. Mercator) suits navigation, topography, and weather maps — anything where measuring angles or reading true shapes matters. Equal-area (e.g. Gall-Peters) suits thematic and density mapping, where the reader compares the sizes of countries or continents — the book's examples are refugee density and cropland in production. (pp. 60–62)